

FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION
Detailed Marking Report — Mathematics Exam-17

Seemab · Class 9 (Science Group) · Units 1, 2, 4, 5, 7 · Attempted 7 September 2026 · deep-check verified

43.5 / 50
TOTAL

87.0%
PERCENTAGE

A
GRADE

6 / 10
SECTION A

23.5 / 24
SECTION B

14 / 16
SECTION C

SECTION A — every question (6 / 10)

#	Question	Hers	Key	Mark	Note and source in her textbook
1	Simplify $(3^4 \times 3^{-2}) \div 3^{-3}$	B	B	1	Unit 1, §1.5.3 Properties of Exponents — product rule ($a^m \cdot a^n = a^{m+n}$) and quotient rule ($a^m \div a^n = a^{m-n}$).
2	$\sqrt{48} + \sqrt{27}$	D	C	0	Unit 1, §1.4.3 Properties of Radicals — Product Rule for Radicals, Example 6. Watch out: $\sqrt{48} + \sqrt{27}$ is NOT $\sqrt{75}$ — you can never add the numbers under two different radicals
3	$\log_5(1/25)$	A	A	1	Unit 2, §2.2 Definition of Logarithm ($\log a = y \Leftrightarrow a^y = b$).
4	If $\log 2 = 0.3010$, $\log 128 = ?$	D	D	1	Unit 2, §2.4 Laws of Logarithm — power law, $\log(a^n) = n \cdot \log a$; Exercise 2.2-style "evaluate using given log values".
5	Factorise $x^2 - 14x + 49$	B	B	1	Unit 4, §4.1 Factorization Types — $a^2 \pm 2ab + b^2 = (a \pm b)^2$.
6	HCF of $15a^3b^2$ and $20a^2b^3$	C	C	1	Unit 4, §4.2 HCF by Factorization (monomial HCF using lowest common powers).
7	$ -8 + 3 $	D	A	0	Unit 5, §5.3.1 Absolute Value. Watch out: the bars in $ -8 $ are NOT brackets — they mean "distance from zero," which is never negative. The book states this directly in §5
8	If $-3 \leq 2x - 1 \leq 5$, $x \in ?$	B	D	0	Unit 5, §5.4 Compound Inequalities (operating on all three parts of a "sandwiched" inequality together). Watch out: whatever you do (add, subtract, divide), you must do it
9	Distance (0,0) to (9,12)	B	B	1	Unit 7, §7.1 The Distance Formula, Example 2.
10	Mid-point of A(-5,3), B(1,-7)	illegible	C	0	Wrote a coordinate instead of a letter; first number overwritten and illegible. Ruled WRONG by Atif, 9 Sep. Unit 7, §7.2 Mid-Point of Two Given Points, Example 5. Watch out: the mid-point formula AVERAGES the coordinates — it divides the sum by 2, it does not just add them. (-5

Red rows lost the mark; amber rows are judgement calls. Every letter was re-read from the scan and cross-checked by an independent deep-check pass.

43.5

out of 50 · 87%

94%

Sections B + C

60%

MCQs

The short version. Seemab scored 43.5 out of 50. But the single number hides the real story: her written work was excellent at 37.5/40, while her multiple choice collapsed to 6/10. Those are two different problems, and only one of them is about mathematics.

In the written sections she handles negative numbers correctly again and again, and on one question she even caught her own arithmetic slip mid-working and corrected it. Under multiple-choice time pressure, with nothing written down, she made errors she demonstrably does not make when she shows her steps. The fix is not more topic revision. It is making her do rough working for MCQs instead of answering them in her head.

The best news: the two questions she had never encountered in any previous paper were the two she scored full marks on.

Section A — Multiple Choice (6/10)

Q	Topic	Hers	Correct	What happened
1	Laws of exponents	B	B	Correct
2	$\sqrt{48} + \sqrt{27}$	D (75)	C ($7\sqrt{3}$)	Added the numbers under two different radicals: $48+27=75$. Not a sign error — a product-rule-for-radicals gap (Unit 1 §1.4.3). Needed $48=16 \times 3$ and $27=9 \times 3$ first.
3	$\log_5(1/25)$	A	A	Correct
4	$\log 128$	D	D	Correct
5	Perfect-square trinomial	B	B	Correct
6	HCF of monomials	C	C	Correct
7	$ -8 + 3 $	D (-5)	A (11)	$-8+3 = -5$: she read the bars as brackets. Note across all 16 earlier papers she was only ever asked to <i>solve</i> absolute-value equations, never to evaluate one.
8	Compound inequality	B ([1,3])	D ([-1,3])	Dropped the minus on the lower bound. She does this correctly in Section B.
9	Distance formula	B	B	Correct
10	Midpoint	wrote a coordinate pair, not a letter		Unresolved. She wrote "(?, -2)" and the first number is overwritten with solid ink. Checked at 600 dpi under four binarization thresholds; no stroke structure survives. Not guessed.

Section B — Short Questions (23.5/24)

Q	Topic	Mark	Comment
2(i)	Laws of exponents	4/4	$16^{3/4} \times 16^{-1/4} = 4$; second part reduced cleanly to $1/(x^2y^4)$
2(ii)	Logarithms	4/4	Expansion correct; $\log 0.9 = 2\log 3 - 1 = -0.0458$, negative result handled correctly
2(iii)	Find Q from HCF and LCM	4/4	Never set in any of the 16 earlier papers. Used $Q = (\text{HCF} \times \text{LCM})/P$, factorised, cancelled, long-divided to $Q = x^2 - 5x + 4$ with zero remainder. Fully correct.
2(iv)	Sum of three fractions	4/4	Factorised all three denominators, built the LCD, collapsed the numerator to $3x^2$. Answer $3x^2/((x^2-1)(x^2-4))$
2(v)	Absolute value + inequality	3.5/4	Roots 6 and -7 correct; set-builder notation correct; $-3 < x \leq 4$ with the negative bound right. -0.5: the question said "verify both solutions" and no substitution check appears anywhere.
2(vi)	Right-angled triangle from coordinates	4/4	Also never set before. $AB=5$, $BC=10$, $AC=\sqrt{125}$, Pythagoras confirmed, right angle correctly located at B. She initially miswrote $(-5)^2$ as $(-15)^2$, caught it herself and corrected to 25.

Section C – Long Questions (14/16)

Q	Topic	Mark	Comment
3(i)(a)	Product of algebraic fractions	4/4	Full factorisation and cancellation to $x/(x+2)$
3(i)(b)	Square root by division	3.5/4	$3x^2-2x+1$ with zero remainder, arithmetic fully correct. -0.5: omitted the $\pm$. Her own textbook (Unit 4, Examples 29 and 30) ends these with " $\therefore$ Square root = $\pm(\dots)$ ", so this is her book's convention, not an outside nitpick.
3(ii)(a)	Distances	4/4	$PQ=5$, $QR=5$, $PR=10$
3(ii)(b)	Collinearity	1.5/2	-0.5: showed $5+5=10$ but never wrote "hence P, Q, R are collinear", and labelled the points $AB/BC/AC$ instead of $PQ/QR/PR$ (carried over from 2(vi))
3(ii)(c)	Midpoint	1/2	-1: computed (4,6) correctly but never stated it coincides with Q, which was half of what the question asked

What this tells us

1. She is strong where it counts.

37.5/40 on written work. The two questions built specifically to close gaps her school exam exposed — finding a polynomial from its HCF and LCM, and proving a right angle from coordinates — both scored full marks despite being question types she had never met.
2. The MCQ gap is a method problem, not a knowledge problem.

She was previously diagnosed as weak on negative numbers. Her written work disproves that: $-3 < x \leq 4$, $(-3)^2$ and $(-4)^2$, $\log 0.9 = -0.0458$, $7-(-3)=10$, and a self-caught sign correction, all handled correctly. She only slips when answering fast with nothing on paper. **Action: she should do rough working for MCQs rather than mental arithmetic.**
3. A recurring habit is quietly costing her marks.

Three of her four deductions are for not writing the concluding sentence: no verification on 2(v), no "hence collinear", no "which is Q". The same thing appears in exam-10 (quadrant not stated, -1) and exam-13 (quadrilateral not classified, -1). She is doing the mathematics and then not finishing the sentence. **This is worth roughly 2 to 2.5 marks a paper and takes five seconds to fix.**

Revision priorities

- Product rule for radicals** (Unit 1 §1.4.3). $\sqrt{48}+\sqrt{27}$ is not $\sqrt{75}$. Simplify each radical first, then add like terms. This is a genuine rule gap, distinct from the sign slips.
- Evaluating absolute values**, not just solving equations with them. $|-8| = 8$, because the bars mean distance from zero. Her book states it directly in Unit 5 §5.3.1: " $|5| = 5$ and $|-5| = 5$ ".
- Always write the final sentence.** If a question says "show that", "hence", or "verify", the conclusion must be written even when the working already proves it.
- Rough working for MCQs.** The single highest-value habit change available to her right now.

Marking was independently deep-checked: all 18 answers re-derived from the question paper rather than the answer key, and every mark reconciled. Both passes reached 43.5/50 independently. Q10 remains illegible and was not guessed — if it reads $(-2,-2)$ her total is 44.5/50 (89%).